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The Diffusive Mass Transport in Molten Semiconductors (DTIMS) program proposes to experimentally and theoretically investigate the effect of buoyancy driven convection on the measured diffusion coefficient and to investigate the use of an applied magnetic field as a countermeasure to suppress this convection. If successful, this program will greatly reduce the number of samples required to be processes on orbit in the future.
The matrix material will be molten germanium (Ge) and result in 36 diffusion coefficients being measured. The variables to be investigated are 1) dopants, antimony (Sb), and gallium (Ga); 2) capillary diameter, 1.0, 2.0, and 3.0 mm diameters; 3) applied axial magnetic fields, 0.5, 1.0, and 2.0 Tesla; and 4) temperature, 950, 1000, and 1050 C.

The shear cell technique will be used to obtain a measured diffusion coefficient. In this technique two initially separated capillary columns of differently doped materials are melted. The melted columns are then brought into contact with each

Task Description:	other and allowed to diffuse for a specified period of time. The molten columns are then 'sheared' into small segments, which are then solidified. After removal of the solid segments from the shear cell, the average dopant concentration for each segment is measured and plotted against distance. An appropriate solution to Fick's 2nd law is then used to calculate the diffusion coefficient. The objectives of this proposed research are: 1) Investigation of Dmeasured vs. Capillary Diameter For this objective, Sb-doped Ge will be used to investigate the capillary diameter dependence of Dmeasured proposed by Alboussiere et al., where H is the capillary diameter. The hypothesis is that the convective interference should scale as H^8. If sufficiently accurate diffusion coefficients can be measured as a function of capillary diameter, then it may be possible to extrapolate the data to a zero capillary diameter and thus, determine the Dintrinsic value. 2) Investigation of Dmeasured vs. Magnetic Field For this objective, Sb-doped Ge will be used to investigate the proposed H^2/B^4 dependence of Dmeasured proposed by Alboussiere et al., where H is the capillary diameter and B is the applied axial magnetic field measured in Tesla (T). The hypothesis is that in the asymptotic limit of an infinitely large magnetic field, the convective flow velocity will tend to zero and Dmeasured = Dintrinsic. 3) Investigation of Dintrinsic vs. Dopant For this objective, gallium (Ga), will be used to investigate the dependence of Dintrinsic on dopant size (Ga is smaller than Sb) and dopant type (n-type (Ga) and p-type (Sb)). These data will be of great benefit to atomistic modeling efforts exploring the 'residual covalency' and its effect on the transport properties near the melting point. Experimental verification of these models will hopefully lead to a fundamental understanding of the structure of liquids. 4) Investigation of Dintrinsic vs. Temperature For this objective, Sb-doped Ge will be used to investigate the dependence of Dintrinsic on temperature. Based on Prof. Stroud's ab initio modeling, it is suggested that the strong covalent bonds of the germanium solid begin to form while still in the liquid state. Viscosity measurements indicate that this begins at approximately 1000 C. Below this temperature, the liquid begins to 'stiffen' and the transport processes are retarded. Above this temperature and the 'residual covalency' has dissipated and the liquid behaves as a simple metal. The fundamental mechanisms of mass diffusion in the liquid state are still unclear to the degree necessary for the prediction of diffusion of one species into another or even within itself. This observation is especially true with respect to the dependency of diffusion mechanisms on temperature and on concentration levels, as well as, on the dopant type. Present estimates of the diffusivity in molten semiconductors can typically provide an order of magnitude estimate only, without any information on their dependency on concentration levels and types, and on temperature and temperature gradients. The availability of these data is of paramount importance for practical reasons as well. The relevancy of numerical modeling for the analysis and design of ground-based and space-based experiments is directly dependent upon the accuracy of the fundamental material properties used in simulations. These data are also important for the correct characterization and interpretation of experimental results from ground-based and space-based experiments. The subject of how a mass of one species diffuses through a matrix of another species is, at the same time, both a very old and a very new research area. That this area can encompass the small, i.e. movement of electrons in a plasma, to the very large, i.e., the depletion of the global ozone layer, merely serves to emphasize the fundamental aspects of this subject. Most manufacturing technologies at some stage, rely on diffusion processes in the solid, liquid or gas. The need for precise measurements of the diffusion coefficients in molten semiconductors has been repeatedly pointed out. These data are required both to interpret the experimental results from previous space-based (and earth-based) experiments and also to optimize newly envisioned experiments. Difficulties in experimental techniques and theoretical interpretations are cited for the lack of these data. This is a comprehensive program which addresses both of these issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The structure of the liquid state is one of the last frontiers on modern science. Currently, there are no models of the liquid state that can satisfactorily predict the transport phenomena observed in liquids. These measurements will greatly contribute to these efforts.
Task Progress:	Close out of this program continues and several papers are being prepared for submission to archival journals.
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